



APEX SUMMIT

CHICAGO 2026

Parsl Code of Conduct

In the interest of fostering an open and welcoming environment, we as contributors and maintainers pledge to making participation in our project and our community a harassment-free and bullying-free experience for everyone, regardless of age, body size, disability, ethnicity, sex characteristics, gender identity and expression, level of experience, education, socio-economic status, nationality, personal appearance, race, religion, or sexual identity and orientation.

Examples of behavior that contributes to creating a positive environment include:

- Using welcoming and inclusive language
- Being respectful of differing viewpoints and experiences
- Gracefully accepting constructive criticism
- Focusing on what is best for the community
- Showing empathy towards other community members
- Respecting the work of others by recognizing acknowledgment/citation requests of original authors
- Being explicit about how we want our own work to be cited or acknowledged

This meeting will follow the same Code of Conduct.

Issues: contact Dan Katz (dskatz@illinois.edu)

https://github.com/Parsl/parsl/blob/master/CODE_OF_CONDUCT.md

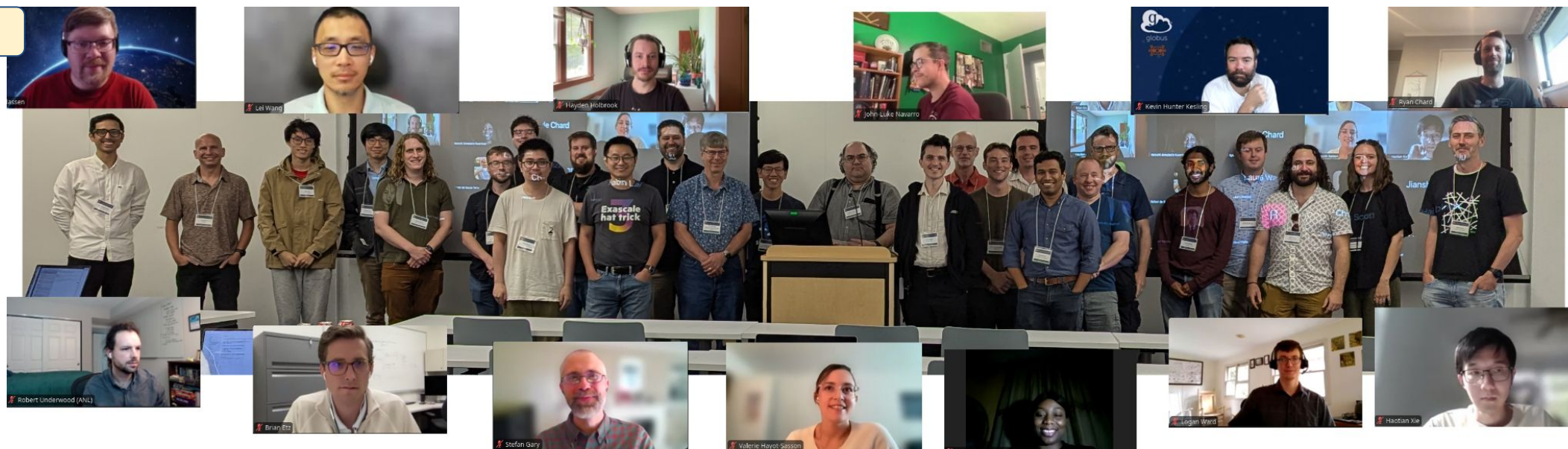


Welcome to the 8th Annual ParslFest!

2019



2025



Thanks to funding agencies and project partners



NSF: 2209919 (UChicago), 2209920 (UIUC), 1550588 (UChicago/UIUC), 1550476 (Notre Dame), 1550475 (Colorado State), 1550562 (Northern Arizona), 1550528 (College of New Jersey)
Chan Zuckerberg Initiative



NSF: 2004894 (UChicago), 2004932 (UIUC)

LSST Dark Energy Science Collaboration (DESC)

Argonne LDRDs

- 2022-0230 Productive Exascale Analysis Workflows for Numerical Cosmology
- 2021-0152 Creating a Robust and Scalable Framework for On-demand Analysis and AI-based Experiment Steering
- 2019-0217 Establishing a Usable, Scalable, and Reproducible Computational Ecosystem for Dark Energy Science

DOE ECP PRJ1008564 ExaWorks project

DOE DE-NA0003963 Center for Exascale-enabled Scramjet Design (CEESD)

Discovery Partners Institute (DPI): Airborne-Satellite-AI-HPC integrative framework (ASAI)

Center for Molecular Modeling, Ghent University

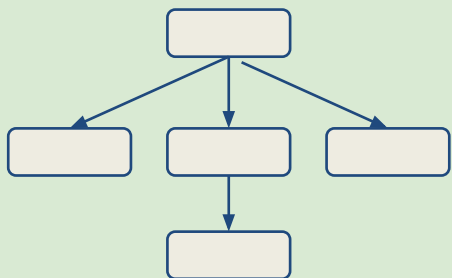
DOE ModCon: Transformational AI and Data





Workflows

Structured Execution

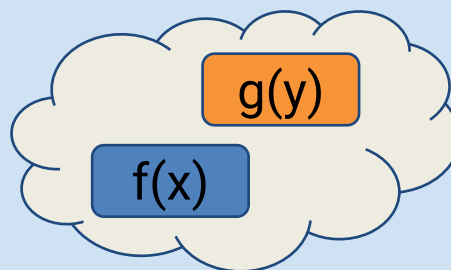


- Compute graph
- Parallel tasks
- Structured logic

Scripts to
Functions

Functions

Remote Computation

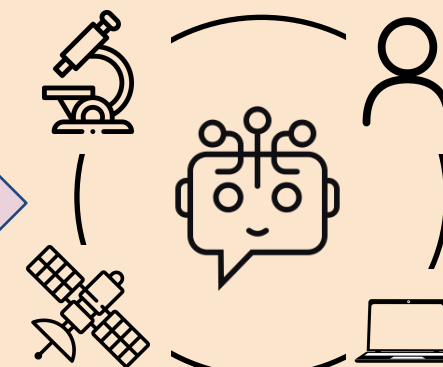


- On-demand exec
- Elastic resources
- Central orchestration

Functions
to Agents

Agents

Autonomous Systems



- Persistent agents
- Real-time adaption
- Decentralized exec

Parsl: parallel programming in Python

Apps define opportunities for parallelism

- Python apps call Python functions
- Bash apps call external applications

Apps return “futures”: a proxy for a result that might not yet be available

Apps run concurrently respecting dataflow dependencies. Natural parallel programming!

Parsl scripts are independent of where they run.
Write once run anywhere!

```
pip install parsl
```

```
@python_app
def hello():
    return 'Hello World!'

print(hello().result())
```



Hello World!

```
@bash_app
def echo_hello(stdout='echo-hello.stdout'):
    return 'echo "Hello World!"'

echo_hello().result()

with open('echo-hello.stdout', 'r') as f:
    print(f.read())
```



Hello World!

Babujji et al. Parsl: Pervasive Parallel Programming in Python. HPDC. 2019

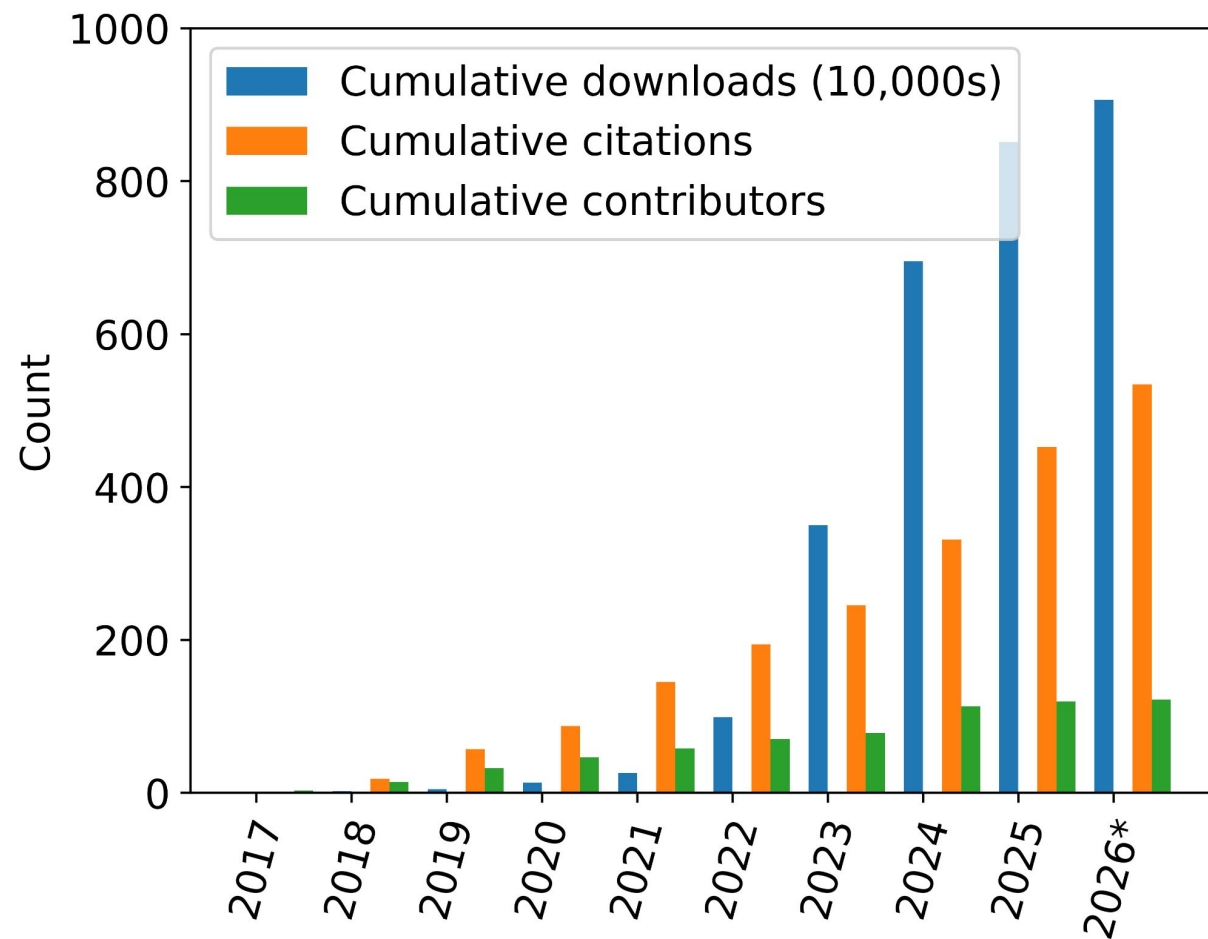
Parsl update

2026 highlights

- Continued robustness / sustainability work
- Memoizer/checkpoint plugins
- Log configuration
- HTEX work for Globus Compute
- RP executor

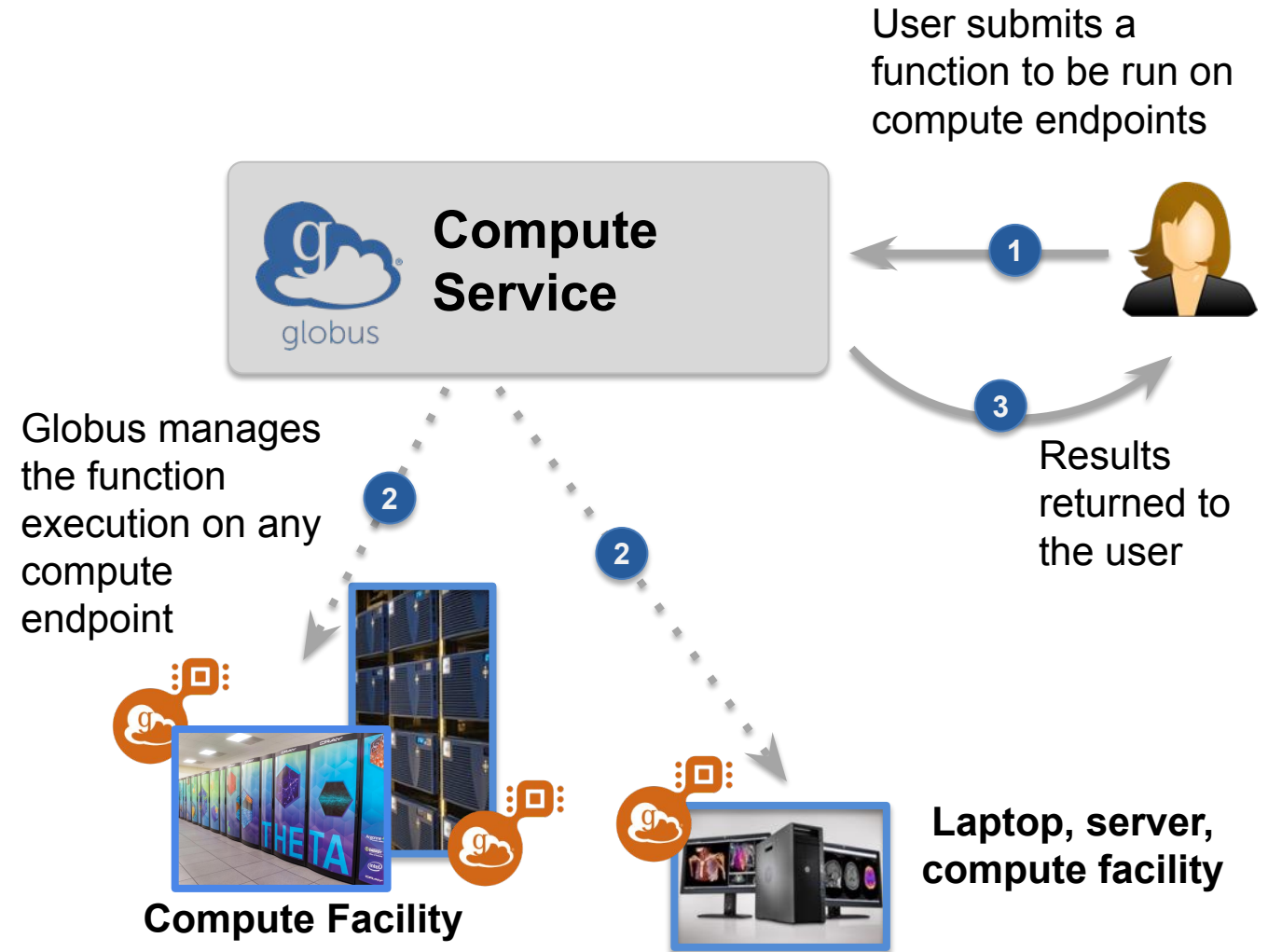
Ongoing work

- Maintenance / bug fixing
- Enhancements to support higher layers (e.g., observability, monitoring, task management)



Managed FaaS compute ...on any system

- Support use of Python for functions
- Fire and forget function execution
- Federated authentication, and local access control
- Uniform interface to various compute resources



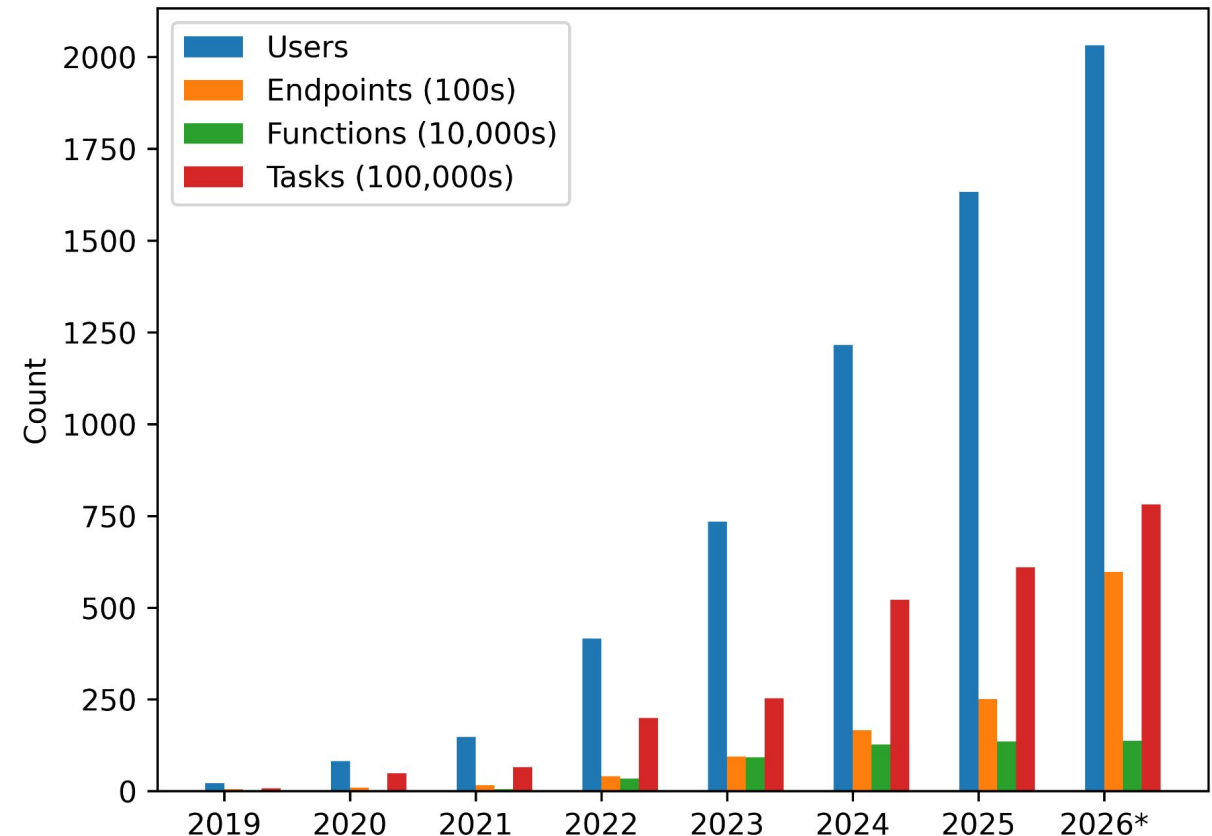
Globus Compute Update

2026 highlights

- Multi-user enhancements and deployments
- Scalability and robustness enhancements
- Web app enhancements including function registration and task submission
- MCP and AI interfaces (HPC Bridge)

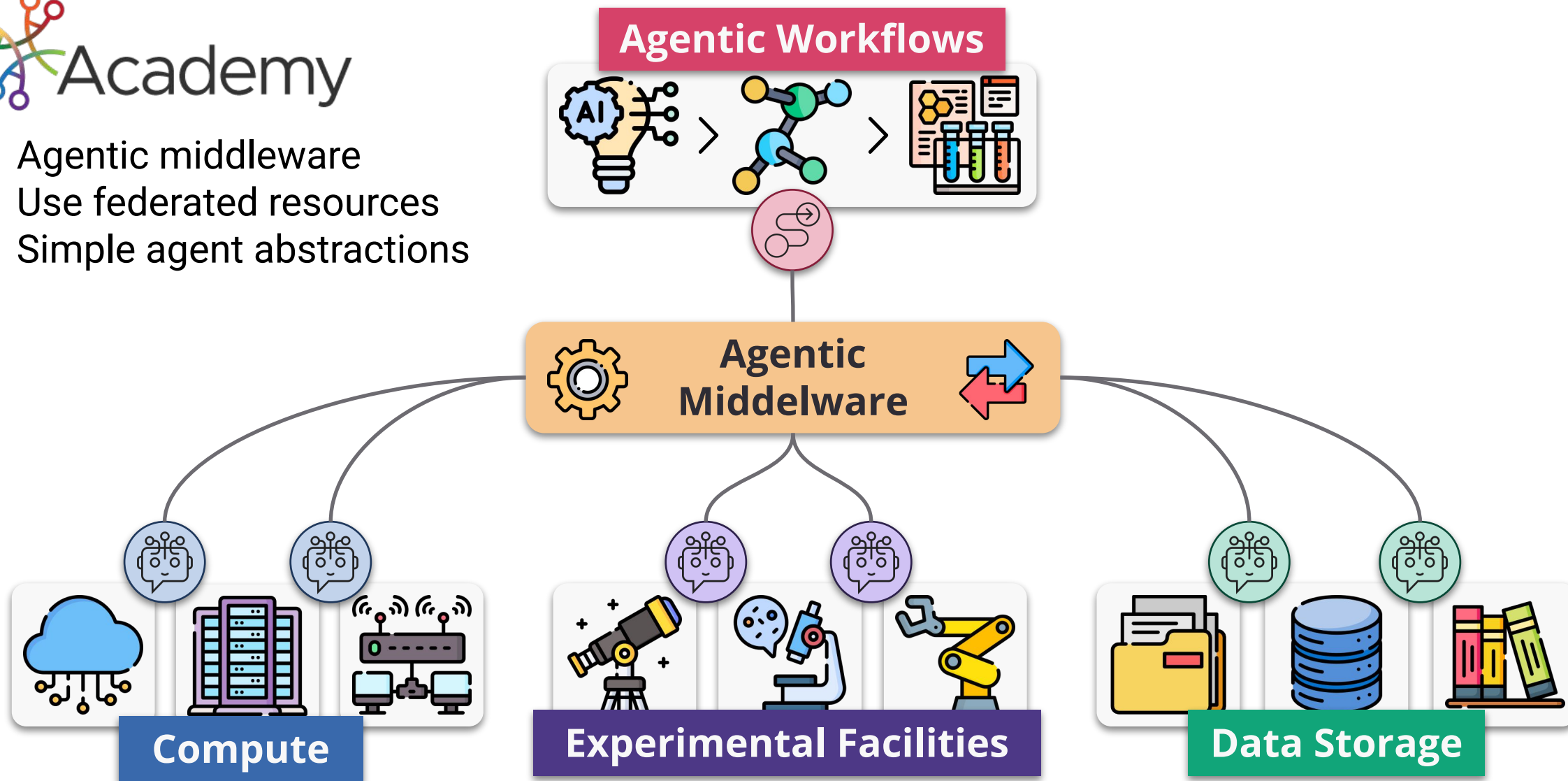
Ongoing work

- Service rearchitecture for performance/scalability
- Environment management
- Fine-grain task management





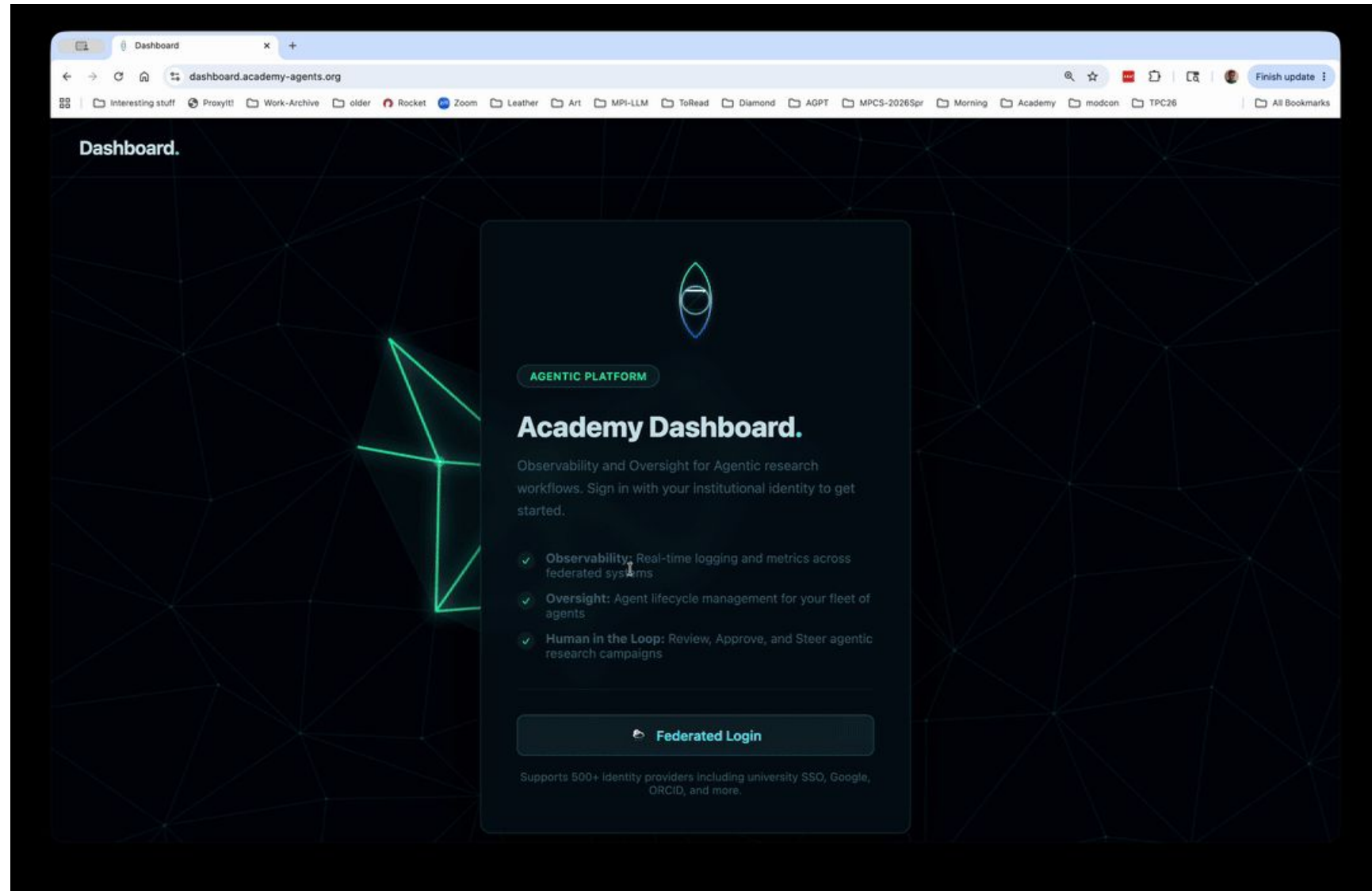
- Agentic middleware
- Use federated resources
- Simple agent abstractions



Kamatar, Pauloski et al. Empowering Scientific Workflows with Federated Agents, IPDPS, 2026

Academy update

- Productionizing managed exchange
- Fine-grain agent authorization model
 - E.g., mailbox sharing
- Managed control plane
- Reputation service
- Common memory services





APEX SUMMIT

CHICAGO 2026

Monday, September 14

1:00 pm - Welcome!

1:10 pm - Session 1

2:10 pm - Break

2:30 pm - Session 2

3:40 pm - Break

4:00 pm - Session 3

5:15 pm - Day 1 Closing

5:30 pm - Dinner (Nella Pizza e Pasta)

Tuesday, September 15

8:30 am - Breakfast

9:00 am - Session 4

10:10 am - Break

10:25 am - Session 5

11:35 - Break

11:50 - Session 6

12:50 - Lunch

Nella Pizza e Pasta

